Answers

Problem 1

(a) $\forall x \in \mathbb{R},\ x^2 + 1 > 0$.

(b) $\exists n \in \mathbb{N}$ such that $n^2 > 100$.

Problem 2

Continuity at $x = 1$ requires

$$\lim_{x \to 1-} f(x) = f(1).$$

Here

$$\lim_{x \to 1-} (x^2 + a) = 1 + a, \qquad f(1) = 2.$$

Hence $1 + a = 2$, so

$$a = 1.$$

Problem 3

The function $f(x) = x^3 + x - 1$ is continuous on $[0, 1]$. Also

$$f(0) = -1 < 0, \qquad f(1) = 1 > 0.$$

By the intermediate value theorem, there exists $c \in (0, 1)$ such that

$$f(c) = 0.$$

Thus $x^3 + x - 1 = 0$ has at least one solution in $(0, 1)$.

Problem 4

(a) By the product rule,

$$f'(x) = 2x \sin x + (x^2 + 1) \cos x.$$

(b) By the product rule and chain rule,

$$g'(x) = (\log x)^2 + 2 \log x.$$

Problem 5

This is a 0/0 form, so by l'Hopital's theorem,

$$\lim_{x \to 0} \frac{\cos(\sin x) - 1}{x} = \lim_{x \to 0} \frac{-\sin(\sin x) \cos x}{1} = 0.$$

Problem 6

For $f(x, y) = x^3 y^2 + e^{xy}$,

$$\frac{\partial f}{\partial x}(x, y) = 3x^2 y^2 + y e^{xy},$$

and

$$\frac{\partial f}{\partial y}(x, y) = 2x^3 y + x e^{xy}.$$

Problem 7

(a) For $f(x) = \sin x$ at $a = \pi/2$,

$$f(a) = 1, \quad f'(a) = 0, \quad f''(a) = -1, \quad f'''(a) = 0.$$

Therefore the third degree Taylor polynomial is

$$T_3(x) = 1 - \frac{1}{2}\left(x - \frac{\pi}{2}\right)^2.$$

(b) Since every derivative of e^x is e^x and $e^0 = 1$,

$$T_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!}.$$

Problem 8

Euler's formula gives

$$\cos w = \frac{e^{iw} + e^{-iw}}{2}.$$

Putting $w = ix$ gives

$$\cos(ix) = \frac{e^{-x} + e^{x}}{2}.$$

Choice Problem A

Take

$$z = i\log(2+\sqrt{3}).$$

Then

$$\cos z = \cos\left(i\log(2+\sqrt{3})\right) = \cosh\left(\log(2+\sqrt{3})\right).$$

Since

$$\cosh t = \frac{e^t + e^{-t}}{2},$$

with $t = \log(2+\sqrt{3})$ we get

$$\cosh t = \frac{2+\sqrt{3}+\frac{1}{2+\sqrt{3}}}{2} = \frac{2+\sqrt{3}+2-\sqrt{3}}{2} = 2.$$

Thus $z = i\log(2+\sqrt{3})$ is one solution.

Choice Problem B

Let

$$\phi(x) = 1 + \sin\left(\frac{\pi x}{4}\right) - x.$$

On $[0,2]$,

$$\phi'(x) = \frac{\pi}{4}\cos\left(\frac{\pi x}{4}\right) - 1 \le \frac{\pi}{4} - 1 < 0.$$

Also $\phi(2) = 0$, so $\phi(x) \ge 0$ for $0 \le x \le 2$.

Since $a_0 = 0$ and $0 \le a_n \le 2$ implies

$$0 \le a_{n+1} = 1 + \sin\left(\frac{\pi a_n}{4}\right) \le 2,$$

the sequence is bounded above by 2. Moreover,

$$a_{n+1} - a_n = \phi(a_n) \geq 0,$$

so it is increasing. Therefore it converges. If its limit is L, then

$$L = 1 + \sin\left(\frac{\pi L}{4}\right).$$

The function ϕ is strictly decreasing on $[0, 2]$ and $\phi(2) = 0$, hence $L = 2$.

Choice Problem C

Put $x = \tan a$ and $y = \tan b$. Then $x, y \in [0, 1)$. It is enough to prove

$$\arctan\sqrt{xy} \leq \frac{\arctan x + \arctan y}{2},$$

because tan is increasing on $[0, \pi/2)$.

Define

$$F(t) = \arctan(e^t).$$

Then

$$F''(t) = \frac{e^t(1 - e^{2t})}{(1 + e^{2t})^2}.$$

For $x, y \in [0, 1)$, write $x = e^u$, $y = e^v$ with $u, v \leq 0$ when $x, y > 0$. On $(-\infty, 0]$, $F''(t) \geq 0$, so F is convex. Hence

$$\arctan\sqrt{xy} = F\left(\frac{u+v}{2}\right) \leq \frac{F(u) + F(v)}{2} = \frac{\arctan x + \arctan y}{2}.$$

If $x = 0$ or $y = 0$, the left side is 0 and the claim is immediate. Thus

$$\sqrt{\tan a \, \tan b} \leq \tan\left(\frac{a+b}{2}\right).$$

Choice Problem D

The displayed expression appears to intend the limit as $x \to 0$. Using Taylor expansions at 0,

$$e^{-x} = 1 - x + \frac{x^2}{2} - \frac{x^3}{6} + \frac{x^4}{24} + O(x^5),$$

and

$$\cos(\sin x) = 1 - \frac{x^2}{2} + \frac{5x^4}{24} + O(x^6).$$

Therefore

$$e^{-x} - \cos(\sin x) + \frac{x^3}{6} = -x + x^2 - \frac{x^4}{6} + O(x^5).$$

The first nonzero term is $-x$. Hence the required natural number is

$$n = 1,$$

and the limit is

$$\lim_{x \to 0} \frac{e^{-x} - \cos(\sin x) + x^3/6}{x} = -1.$$

Choice Problem E   The blanks are

$$\begin{aligned}
[1] &= \text{Brook Taylor},\\
[2] &= \text{James Gregory},\\
[3] &= \text{Maclaurin},\\
[4] &= \frac{f^{(n)}(0)}{n!}x^n.
\end{aligned}$$

Thus the expansion is

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n.$$

Choice Problem F   For e^x at $a = 0$,

$$T_n(x) = \sum_{k=0}^{n} \frac{x^k}{k!}.$$

By Taylor's theorem with Lagrange remainder, for some $c \in (0, 1)$,

$$e - T_n(1) = \frac{e^c}{(n+1)!}.$$

Since $e^c < e < 3$, if $n > 3$, then

$$0 < e - T_n(1) < \frac{3}{(n+1)!} < \frac{1}{n!}.$$

Assume for contradiction that $e = p/q$ with $p, q \in \mathbb{N}$. Choose $n > 3$ with $n \geq q$. Multiplying

$$0 < e - T_n(1) < \frac{1}{n!}$$

by $n!$ gives

$$0 < n!\,(e - T_n(1)) < 1.$$

But $n!e = n!p/q$ is an integer because q divides $n!$, and

$$n!T_n(1) = \sum_{k=0}^{n} \frac{n!}{k!}$$

is also an integer. Hence $n!(e - T_n(1))$ is an integer strictly between 0 and 1, which is impossible. Therefore e is irrational.